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An expectation-based metric for NFL field goal kickers

Abstract: The standard metric for American football field goal kickers is simply the percentage of attempts successfully converted. Due to variance in distance of attempts and other conditions (weather, altitude, defense, etc.), we argue that field goal percentage is an insufficient measure of kicker performance. Using three seasons of NFL data, we construct a multivariate logistic regression model for the success probability of a given attempt. This leads naturally to metrics in which a kicker's performance is compared to model expectations, if a replacement-level player was attempting the same kicks. Player salaries correlate only weakly with our measures of field goal kicking success. We find that those kickers selected to the Pro Bowl and All-Pro teams were rather mediocre by our metrics, over the seasons studied. The relative difficulty of kicking in various stadiums is also considered. Finally, we discuss the degree to which field goal kicking is a skill that can be maintained over multiple seasons.

Keywords: kicker; field goal; football; replacement player; expectation.

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1 Introduction

Field-goal kicking in the National Football League (NFL) is often a critical part of the game. As compared to players at most other positions, a placekicker's outcomes can be more easily objectively rated, because there are fewer interactions with other players to consider. For example, a quarterback's statistics depend on the play of his receivers and their defenders, his offensive line and the opposing pass-rushers, etc. By comparison, a kicker largely works alone, depending primarily on the snapper and holder. Because the typical snap-to-kick time is 1.2–1.3 s

(Crossman 2013), less than half of the average snap-to-pass time for a quarterback (Elsayed 2012), blocked kicks are less likely than quarterback sacks, despite having more defensive rushers on a kick.

Kickers are commonly evaluated based on field goal percentage (that is, the percentage of attempts that are successfully converted), or simply the number of kicks made. However, some kickers more frequently attempt long field goals than others, and weather factors (such as cold temperatures, strong winds, rain, or snow) may also impact the chance of a particular kick being made. For these reasons, field goal percentage is an inadequate metric to assess the quality of kickers.

Value-added and efficiency-based metrics are increasingly being used for player evaluation in a variety of sports. Perhaps the best known is wins above replacement player (WARP, or just WAR), which attempts to measure a baseball player's overall contribution to his team, measuring his contribution in team wins added, as compared to an ordinary *replacement* player at the same position, who could be brought off the bench or called up from the top-level minor league team. Other value-based statistics include effective field goal percentage in basketball, EPA (expected points added) in football, and adjusted plus/minus in hockey (and basketball). We believe that such an approach is also potentially useful in evaluating field goal kickers.

The remainder of this paper is organized as follows. Section 2 gives a brief history of field goal kicking and summarizes prior research. Section 3 describes the categories of data that we compiled for each data point. Section 4 discusses a simple distance-only logistic regression model and applies it to NFL stadiums. Sections 5 and 6 consider the impact of other potential factors. Section 7 presents the full model and revisits the issue of particular stadiums. Sections 8 and 9 apply the full model to assessing individual kicker performance. Finally, Section 10 discusses limitations of the model and future work.

2 Historical background

For seasons prior to the 1970 NFL/AFL merger, the data regarding distances on NFL field goal attempts is

incomplete, so studying field goal kicking in a scientific manner would have been very difficult before that time. Berry and Berry (1985) ranked kickers using only the distances of their successful and failed field goal and point-after-touchdown (PAT) attempts. Using a logistic model, the authors determined the distance at which each kicker would be expected to have 50% accuracy, and ranked them accordingly. Using data from the 1989 to 1991 seasons, Morrison and Kalwani (1993) did not find significant differences in skill among NFL kickers.

Subsequently, Bilder and Loughin (1998) considered generic field goal probabilities (considering all kickers to be equally skilled) with a variety of explanatory variables, including some related to weather (via binary variables for precipitation and strong winds) stadium (altitude, indoor/outdoor and surface), and game situation (time, score, week of season). In addition, they found strong winds (above 15 mph) and pressure (the potential of a kick to cause a lead change) to be significant. Several variables which are considered important in football's "traditional wisdom," but were not statistically significant in the Bilder study, included altitude, surface (natural grass versus artificial turf), temperature, precipitation, and stadium type (domed versus open-air).

More recently, Football Outsiders (2013) combines all weather-related effects into a single category, classifying stadiums in one of four categories: dome, cold climate, warm climate, or Denver (high altitude). This model considers only stadium category and distance.

The accuracy of NFL field goal kickers has significantly increased over the last several decades, as shown in Figure 1 using a locally-weighted quadratic least-squares regression (LOESS) technique. The percentages of

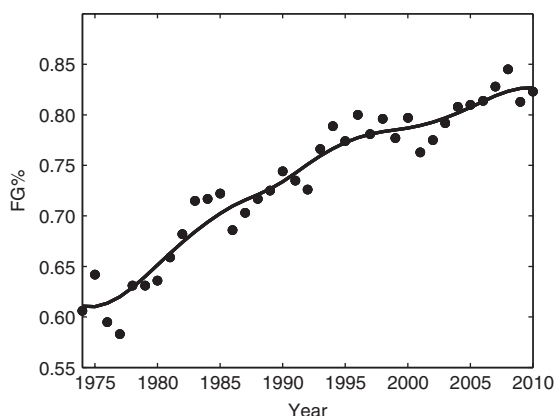


Figure 1 Historical regular-season field goal success rates, with LOESS trend. We present data beginning in 1974, the year in which the goal posts were moved from the goal line to the back of the end zone. This data set is taken from pro-football-reference.com.

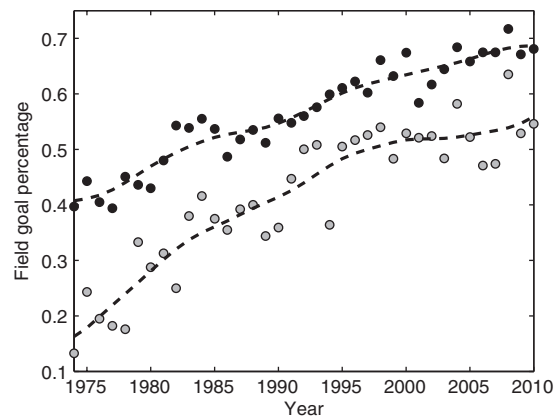


Figure 2 Historical regular-season success rates on field goals of 40+ yards (dark dots) and 50+ yards (light dots), both with LOESS trend.

field goal attempts taken from 40 or more yards, and from 50 or more yards, have remained steady since the 1980s, at about 40% and 10%, respectively. However, accuracy at these longer distances has steadily increased, as shown in Figure 2. A PAT is essentially a 20-yard field goal, and PAT failure rates have decreased from about 5% in the 1980s to around 1% over the last decade. Explanations for these improvements include soccer-style techniques, better instruction regarding kicking form (Maske 2011), improved long-snappers, and strength training (Stapleton 2010). This improvement has continued in recent years, despite a 1999 rule change requiring new "K" footballs (as opposed to broken-in balls) to be used on special teams plays. Kickers believe that the brand new footballs are harder, and do not travel as far (McCallum 1999).

3 Data collection

With the availability of extensive play-by-play data in online sources, we consider every regular-season and post-season field goal attempt over the 2008, 2009, and 2010 NFL seasons. For each kick, the categories of game-related data, taken from Pro Football Reference (2012), are listed below.

- Distance (discrete)
- Week number of the season
- Playoff game or regular season
- Minutes remaining in game (rounded down)
- Quarter (overtime included with 4th quarter)
- Score margin (kicking team perspective)
- Absolute score margin
- Stadium
- Kicker's identity (name, team)
- Opponent (defensive team)

Additionally, associated historical weather data was taken from wunderground.com, in categories below.

- Temperature
- Hourly precipitation rate (water equivalent)
- Dew point
- Humidity
- Wind speed
- Wind direction

Other associated variables include the following:

- Altitude
- Headwind/tailwind speed
- Crosswind speed

Using the above, binary variables are computed:

- Home or away stadium of the kicker
- Dome or open-air stadium
- Surface (natural grass or artificial turf)
- Denver (high altitude)
- Precipitation presence/absence
- Pressure situation (a kick to tie/lead/win, in the last 3 min of regulation or in overtime)
- Number of prior field goal attempts in the game by the kicker

Weather data is generally recorded on an hourly basis, so we use game-time weather for field goal attempts in the first quarter, and move 1 h later for each subsequent quarter (e.g., 3 h after kickoff for field goal attempts in the fourth quarter). For games played in domed stadiums, we assume a temperature of 72°F, relative humidity 50% (leading to a dew point of 52°F), and no wind. The last assumption may be the most dubious, as anecdotal evidence suggests that heating/cooling systems in domed stadiums cause substantial air movement. Also on-field wind direction and speed in an outdoor stadium may be substantially different than that experienced in an adjacent parking lot, let alone at an airport recording station across town.

Based upon physics principles, we expect tailwinds to benefit a kicker, due to decreased air resistance shortening the effective distance of a field goal; by similar logic, headwinds should be detrimental (Gay 2004: 124–125). Crosswinds would also seem to be detrimental, by changing the appropriate lateral “aiming point” by some indeterminate distance. In order to test these hypotheses, we attempt to decompose the wind speed and direction into head/tailwind and crosswind components. In order to do this, we compare the wind direction to the orientation of a given stadium. Using views from Google Maps, we classify each stadium’s longitudinal axis orientation in one

of eight categories, based on a 16-point compass rose: N/S, NNE/SSW, NE/SW, ENE/WSW, E/W, ESE/WNW, SE/NW, or SSE/NNW. Using right-triangle trigonometry, we can compute the component wind speeds in the down-field and lateral directions. One limitation is a lack of information about the goalpost toward which a particular kick was attempted, so we cannot distinguish between headwinds and tailwinds, and thus cannot estimate any related effects. However, because the effect of a crosswind seems likely to be similar regardless of whether the wind blows from the kicker’s left or his right, we have no such issue regarding crosswinds.

Over the 2008–2010 seasons, there were exactly 3000 field goals attempted during the regular season and playoffs, of which 2475 were converted (for a success rate of 82.5%). In taking a crude first look at the data set, we compute correlation coefficients between a binary variable for field goal success and each of the other variables. The highest magnitude correlations were observed for distance ($r=-0.36$) and crosswind (-0.06). These numbers support the idea that distance is, by far, the most important factor and that long field goal attempts are less likely to be successful. However, the relatively low magnitude of the distance correlation leads us to believe that at least one of the following is true: there is substantial “randomness” involved in field goal kicking, individual kickers vary in ability, and/or factors other than distance can (in combination) significantly affect the probability of success. One other notable issue with the data set is that long or difficult field goals may be attempted more often by kickers who are perceived as more skilled, as coaches who are highly confident in their kicker are more likely to call for a difficult field goal attempt, as opposed to punting or running an offensive play. However, we will generate a model for the success probability of a generic NFL kicker converting a field goal attempt under given conditions.

4 Distance-based model

Our first model was a logistic regression model with distance d (in yards) as the sole independent variable. Figure 3 shows this model and the data, which are smoothed by including all attempts within ± 2 yards of a given distance.

$$P(\text{success}) = \frac{1}{1 + e^{-5.8409 + 0.1078d}} \quad (1)$$

Using this model, we can make initial determinations about which stadiums are more or less favorable to

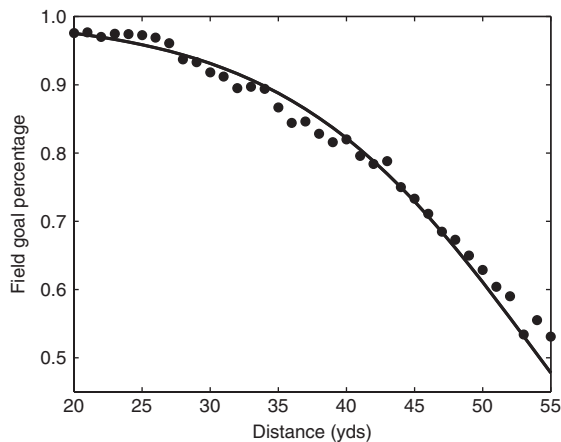


Figure 3 Distance-only logistic regression model and smoothed data (including all attempts within ± 2 yards of the stated distance in the 2008–2010 seasons).

kickers. From the set of field goal attempts taken in a given stadium, we use our model to compute an expected success percentage (ExpFG%), for a generic NFL kicker at a given distance. We compute the difference between the actual and expected success percentages, i.e., $FG\% - \text{ExpFG}\%$. A positive difference indicates aggregate performance that exceeds expectations. The home team's kicker takes about half of the attempts in a given venue, so the validity of the results in a given stadium would be heavily dependent on the assumption that all kickers are equally skilled. To avoid this issue, we will (for now) consider only attempts by visiting kickers. However, because each team hosts their three divisional opponents every year, if a particular division has multiple particularly good or bad kickers, then this could influence the results.

Tables 1 and 2 list the easiest and toughest stadiums for kicking, respectively, according to this distance-only model. Most of those on the first list are domed stadiums or located in warm climates, and those on the second list are outdoor stadiums in cold and/or windy climates. Denver's altitude may explain kickers' success there, as we will see in Section 7. Kicking in Oakland might be difficult due to the Bay Area's high winds, and due to the

Table 2 Toughest stadiums, distance model.

Stadium	FGA	FG%	ExpFG%	Difference
Cleveland	56	71.4%	83.3%	-11.9%*
Arizona ^o	60	71.7%	83.5%	-11.8%*
Chicago	45	75.6%	84.6%	-9.0%
Oakland	58	70.7%	78.1%	-7.4%
New York [†]	53	79.2%	85.9%	-6.7%

^oDomed stadium, [†]Giants Stadium closed after the 2009 season,

* $p < 0.05$.

dirt infield (see Gaines 2012 and Gutierrez 2013), which is an issue during the baseball season on all kicks beyond 45 yards and some in the 35–45 yard range. This leaves Philadelphia and Arizona as clear outliers. The quality of the home team's field goal defense is a potential confounding variable, which we will explore in Section 6. Also, the relatively small sample sizes make randomness significant. The field goal attempts in a particular stadium are independent, but not identically distributed, because of varying distances. For this reason, we use 10,000 simulations (with the model probabilities of individual field goal attempts), rather than a binomial distribution, to estimate the p values referenced in Tables 1 and 2.

5 The modeling process

With “noisy” data, there is a significant risk of *overfitting*, that is, finding trends that represent nothing more than random variation, and developing a model based on those trends. In such a case, the model may fit the data set quite well, yet can be a poor predictor when applied to new data. To avoid this issue, we use 5-fold cross-validation, in which the data set is randomly split into five groups of equal size; a model is built based on the pooled contents of four groups, and validated by how well it predicts on the fifth group. The roles are rotated, with each of the five groups in the validation role once. Because the particular partitioning of the data into five groups can influence the outcome, we repeat this process many times and use the aggregate prediction results to determine whether a variable is predictively useful. See Picard and Cook (1984) for general information on cross-validation techniques.

As a first step, we fit parameters in a multivariate logistic regression model (using the MATLAB software package), using distance and one other variable. For each pair, we ran 100 trials of the aforementioned 5-fold cross-validation procedure. We then compared the errors to those of a distance-only model, to determine whether (and to what degree) the predictions including a second

Table 1 Easiest stadiums, distance model.

Stadium	FGA	FG%	ExpFG%	Difference
Detroit ^o	42	95.2%	81.6%	+13.6%*
San Diego	38	94.7%	83.7%	+11.0%*
Philadelphia	34	91.2%	81.0%	+10.2%
Denver	52	88.5%	78.8%	+9.7%*
St. Louis ^o	43	90.7%	82.9%	+7.8%

^oDomed stadium, * $p < 0.05$.

variable were better than those using distance only. As our metric for error, we use mean squared error (MSE), as shown in equation (2). The observed data values are all either 0 (for an unsuccessful kick) or 1 (for a made field goal), while the model outputs are on a continuous (0,1) scale, thus we are measuring the average of the squares of the model residuals. No individual factor reduced MSE by even 1%, as compared to our distance model; this result reinforces the primacy of distance, and also points to randomness and/or quality differences among kickers.

$$\text{MSE} = \frac{1}{n} \sum_{j=1}^n (\text{Model}_j - \text{Data}_j)^2 \quad (2)$$

Alternatively, we could use our logistic regression model to consider field goal kicking as a classification problem, predicting whether or not a given kick is converted. However, the median length of missed field goals is only 46 yards, and kicks from this distance are successful about 70% of the time, according to our distance-based model. This model suggests that a field goal is more likely to be missed than made only when the distance is at least 55 yards, and such kicks constitute only about 2% of our data set. This is to say that, given the overall success rate of 82.5%, most missed field goals occur in situations where success is more likely than not, so this phenomenon may not be well-suited to a classification approach.

6 Identifying significant factors

Factors which individually improve prediction (in the validation sets) include total wind speed, crosswind speed, temperature, dew point, playoff, head/tailwind, dome, and number of previous kicks. Noting that many of the aforementioned factors do not apply to contests played in domed stadiums, we begin by looking at the differences between kicking indoors and outdoors. In running a model with distance and dome, we estimate the difficulty of an indoor kick is equivalent to that of an outdoor kick from two yards closer. Figure 4 shows the models and (smoothed) data on indoor and outdoor accuracy at various distances. However, this effect might be weather-related (cooler temperatures, wind, etc.), rather than solely due to playing indoors. To investigate this, we fit parameters on a data set including the 674 indoor kicks and the 667 outdoor kicks for comparable weather conditions (which the temperature was between 60 and 84° and the wind speed was <10 mph. Using this crude method of controlling for temperature and wind, we find that the dome variable is neither predictively beneficial

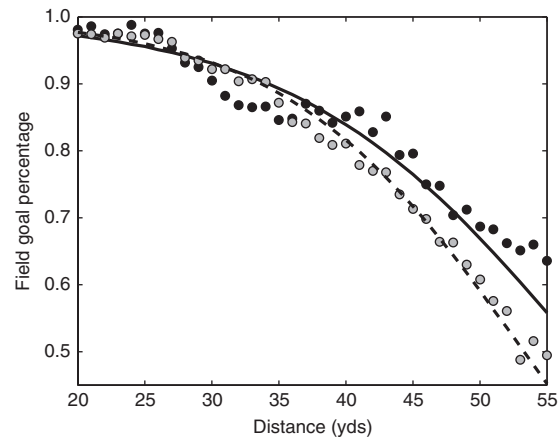


Figure 4 Distance-only logistic regression models and smoothed data for indoor (dark dots, solid line) and outdoor (light dots, dashed line) field goal attempts.

nor statistically significant (at the 0.05 level). Figure 5 shows the smoothed data and combined model for these indoor and mild weather outdoor kicks.

Temperature and dew point are highly correlated (with $r=0.83$) outdoors, so we question whether both factors are needed. Burke (2012) found greater field goal success at warmer temperatures, using some limited empirical data. Based on physics principles, Gay notes that, “Extreme temperatures can affect kicking distances on the level of a few yards... Humidity will have almost no effect on the ball’s range (Gay 2004: 127).” For these reasons, we will initially include temperature, but not humidity. A later check confirmed that neither dew point nor relative humidity were predictively valuable, when added to the final model.

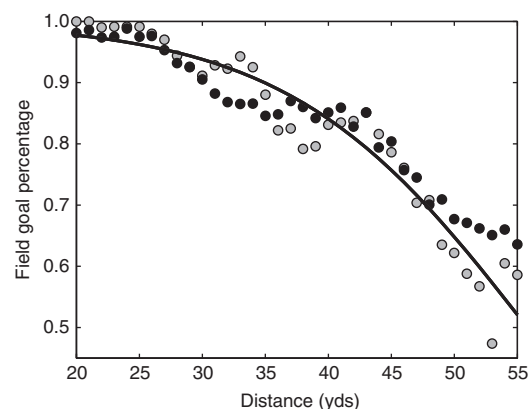


Figure 5 Distance-only logistic regression model and smoothed data for indoor (dark dots) and comparable-weather outdoor (light dots) field goal attempts. While success percentages were greater indoors between 50 and 55 yards, this trend was reversed in the small sample of kicks from beyond 55 yards, as indoor attempts were 0-for-6, while mild-weather outdoor attempts were 6-for-12.

Conventional wisdom suggests that kicking on artificial turf is easier than kicking on grass, but this variable was not helpful in prediction. Also notably missing from our list of predictive variables is precipitation; because of poor traction in wet conditions, rain is believed to hinder kicking, but our predictions were worse when using precipitation as an input, so we will not include it. Only 6% of the kicks in our data set were flagged as occurring during precipitation, and in only 1% of the cases was the precipitation rate 0.1 in/h or higher (of water equivalent), so we have a small sample size of kicks in wet conditions.

Carré et al. (2002) used physics principles to assert that the effects of drag (air resistance) on a kicked football are proportional to the square of the ball's speed. By extension, we considered that wind effects might also be proportional to the square of the wind speed. Using the square of the wind speed reduced error, both with total wind speed and crosswind speed. Among several ways of combining these two variables, we find that the square of their sum, i.e., $(\text{total wind speed} + \text{crosswind component speed})^2$, yields the smallest validation error. In fully quadratic models, the linear term coefficients are nearly zero, so linear effects are insignificant in a quadratic model. Our finding that this variable is slightly more predictive than the square of crosswind alone may indicate shortcomings of a rather crude approach to computing wind speed components. Also, the wind speed and direction inside a large stadium may be different than that present in an adjacent parking lot, let alone an airport several miles away.

Including altitude as a variable was neutral, as the error value in our predictions was almost identical with or without it. Because Gay (2004: 119–123) presents evidence that kicked footballs travel approximately 10% farther at Denver's mile-high altitude than at sea level, we chose to include this variable in our model. Given that Denver is about four times as far above sea level (5280') as the second-highest NFL stadium (Arizona, 1368'), it is not unexpected that a binary variable for Denver works well.

Field goal percentages in the playoffs are well below expectations, with 80 of 106 kicks converted, compared to 88.1 successes that we would expect based on kick distances alone. After adjusting for weather, and even when considering only indoor kicks, the effect remains substantial, equivalent to about five yards, and could be a consequence of psychological pressure in high-stakes contests. Because playoff games involve good teams, which generally have above-average defenses, we seek to determine whether defense quality could explain some of this effect; defense quality also certainly could be important in regular-season kicks. We first attempt to directly measure field goal defense, by comparing the predicted and actual

results of kicks against each team's defense, using only away games (i.e., games in the kicker's home stadium) to reduce extraneous stadium-related effects; however, this measure fails validation tests when included in our model, so we discarded it. We then consider a number of general team-strength variables, including points scored and allowed, scoring margins and ratios, winning percentage, and a binary playoff team variable. Several of these yield good results in validation tests, and the average number of points allowed (per regular-season game) is the most predictive. However, defense quality accounts for only about one-fifth of the playoff effect, so both our defensive measure and a binary playoff variable are included in the final model presented in Section 7.

Defense quality is not an externally measurable factor in a particular field goal attempt, in the same sense as distance or weather conditions, so we must be a bit more cautious about justifying its inclusion. As noted in Table 6, a league-leading defense will effectively lengthen a field goal by about two yards. On a mid-range kick of 40 yards, this reduces the success probability by about 3–4%. Then we expect an excellent defense to allow one less field goal over the course of a season (out of a typical 30 opposing kicks), as compared to an average defense. However, an excellent defense will surrender 100+ fewer points during a 16-game season, so a single field goal is only a small part of that point difference. Because the average-points-allowed measure of defense quality is predictive of field goal outcomes, but not heavily influenced by those outcomes, we include it in our model. To use only already-known data in prediction, an alternative would be to use the season-to-date average points allowed; early in the season, the prior year's average points allowed or the league average might be preferable.

Berry and Wood (2004) considered psychological pressure in the context of important field goals late in close games. They defined a pressure kick as one with the opportunity to tie, lead, or immediately win the game (i.e., when the game is tied or the kicking team is trailing by three or fewer points) in the final 3 min of the fourth quarter or at any point in overtime. While not all overtime field goals are game-ending under current NFL rules, a first-team-to-score-wins ("sudden death") format was still in place during the seasons included in our data set. When we add a binary pressure variable, using the above definition, the aggregate results of the 142 such kicks are only slightly below expectations; the difference is not statistically significant, and adding this factor leads to slightly larger error in the validation sets, so we do not include a pressure factor.

While occasionally an NFL team will retain both a kickoff specialist and a placekicker, most teams keep a single kicker (who handles both roles) on their roster. For this reason, a change of starting kickers normally occurs only either when the kicker is injured or when the team becomes sufficiently dissatisfied with his performance to release him. Over an NFL season, we consider a *starting* kicker to be one who attempted at least 16 field goals (i.e., averaging one per scheduled game) in that season, and we call a non-starter a *replacement* kicker. Our pool of replacement kickers collectively took 6% of the attempts. Admittedly, our starting group includes some players who were not on active rosters at the beginning of the season, and the replacement group has some who lost their jobs early in the season, as well as some good players who suffered injuries. (For example, Jason Hanson's impressive 2010 half-season is in our replacement group, even though he made 12 kicks of 14, including seven from 47 yards or longer, before being injured in mid-season.) In future seasons, we ideally would use opening-day depth charts to determine the 32 starting kickers, and call all others replacement kickers. Note that this is not quite the same as denoting those kickers who played in the first week as starting kickers, as a team's "starter" be injured at the start of the regular season. While imperfect, this is similar to the methodology used by Football Outsiders to identify replacement players at other positions, sorting players by number of attempts in a given category (for example, rushes) and including those who took the "final" 10% of such plays (i.e., those players with the fewest attempts who collectively accounted for 10% of the attempts).

Over the three seasons studied, replacement kickers made only 77.4% of their kicks, compared to 82.8% for starters, even though the average distance of replacement kickers' attempts was a yard shorter. This result is significant at the 0.01 level, and including a binary variable for replacement players improves our model predictions. It is not surprising that kickers with more attempts would outperform the replacement group, because good kickers are more likely to keep their jobs. Differences between starting and replacement kickers are most evident on longer field goals, as shown in Table 3, and are statistically significant only for kicks longer than 40 yards. Also, 40% of attempts by starters are taken from 40 yards or farther, while only 34% of those by replacement kickers come at these longer distances, perhaps indicating that coaches are more willing to allow longer field goal attempts by kickers in whom they have greater confidence. If we fit a logistic regression model using only distance and binary replacement, the expected difference between starting and replacement kickers increases with distance, up to

Table 3 Starting versus replacement kickers at various distances.

Distance (yds)	Starting kickers		Replacement kickers		Difference FG%
	FGM/A	FG%	FGM/A	FG%	
18–29	799/823	97.1%	53/57	93.0%	4.1%
30–39	743/850	87.4%	57/68	83.8%	3.6%
40–49	612/826	74.1%	29/47	61.7%	12.4%
50+	174/311	55.9%	8/18	44.4%	11.5%

50 yards (at which the replacement kicker model has an inflection point), as shown in Figure 6.

An NFL kicker will attempt typically attempt three or fewer field goals (in addition to points-after-touchdown and kickoffs) in a game. However, in 2007, Tennessee Titans kicker Rob Bironas broke the single-game record by making eight field goals, including the game-winner as time expired. As a team's offense drives into the opponent's territory, the kicker warms up by taking practice kicks into a net on the sidelines, even though they may or may not be called on to attempt a kick. We question whether fatigue could have an impact on field goal success; for each field goal attempt, we note how many previous field goals the kicker had already attempted that day. We find performance comparable to expected levels on a kicker's first four attempts of a game, but dramatically substandard results if the kicker had taken four or more prior attempts, as shown in Table 4. We construct a binary fatigue variable, with a kicker defined to be fatigued if he has already attempted at least four field goals in the current game. Using a model with seven previously-discussed factors

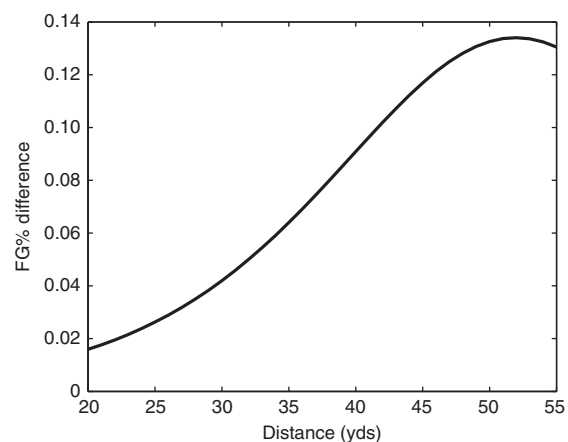


Figure 6 Estimated success probability differential between starting and replacement kickers at various distances, based on a distance-only model.

Table 4 Possible fatigue effects.

Previous FGA	Occurrences	FG%	ExpFG%†	Difference
None	1386	83.5%	83.0%	+0.5%
1	938	82.2%	82.2%	−0.0%
2	459	81.0%	81.3%	−0.2%
3	160	84.4%	84.0%	+0.4%
4+	57	68.4%	81.1%	−12.7%*

†Based on a model including seven previously-discussed factors,

* $p < 0.01$.

(which are listed in Table 5), we find that fatigue is significant at the 0.01 level, despite the small sample size; if we add it to the model, prediction quality is improved, so this becomes the 8th and final factor in our model.

7 A multi-factor model and its implications

From this set of factors, we apply a five-fold cross-validation process, using distance plus all possible subsets of the other factors, to determine which are independently useful for prediction. For each subset of factors, we use 100 random partitions of the data, taking an average of the results. In Table 5, we show the predictive error (MSE) results for three models: a baseline model in which every kick is assumed to have the same probability of success, the distance-only model from Sections 4 and 5, and the “full” eight-factor model discussed in Sections 6 and 7. Note that due to the cross-validation approach, adding irrelevant factors would not decrease the (predictive) error, so there is no inherent advantage to adding more factors. When models with seven of these eight factors were constructed, their rank in predictive importance was (in decreasing order) as follows: distance, wind, temperature, replacement, defense quality, fatigue, playoff, and Denver. The last of these factors is predictively neutral but supported by physical principles, while each of the others is individually significant, even with the others included.

After selecting variables, the coefficients are determined by a regression fit using the full data set (as

Table 5 Comparison of models.

Model	Predictive MSE
Baseline	0.1444
Distance	0.1249
Full	0.1226

opposed to 80% of it in each validation step), and the resulting error values would be slightly smaller than those shown in Table 5, because the coefficients are being determined from all of the data. The general form of our model is shown in equation (3) below, where c_0 is the constant term, and each x_i and c_i represent an input variable and its associated coefficient.

$$P(\text{FGmade}) = \frac{1}{1 + e^{-(c_0 + \sum c_i x_i)}} \quad (3)$$

Table 6 gives the variables and coefficients used in our final full model.

Coefficient ratios show linear relationships among the variables. For example, we estimate that a temperature increase of 13°F effectively shortens a field goal by one yard. The magnitude of this effect is weaker than that found by Burke (2012) in a more cursory study, but perhaps stronger than that expected by Gay (2004: 127). Assuming that the weather is the same, a kick in Denver would have the same success probability as a kick taken from three yards closer in a stadium near sea level; this seems reasonable, given the findings of Gay (2004: 119–123). Table 7 gives the impact of various factors on the effective distance of a kick; the baseline for comparison is a generic starting NFL kicker attempting a field goal during a regular-season game, with an ambient temperature of 72°F and calm winds (or inside a dome).

Because our model involves the square of wind speed, the effects of wind on distance are nonlinear. While a typical 5 mph wind has minimal impact, a crosswind of 15 or 20 mph can effectively lengthen a kick by several yards, as shown in Figure 7.

We can determine the theoretically easiest and most difficult stadiums for visiting kickers by computing the historical average regular-season weather effects for each stadium, accounting for the home team’s defensive quality, and (for Denver) including the effect of altitude,

Table 6 Final model variables and coefficients.

Variable (unit)	Coefficient
Constant term	4.9769
Distance (yds)	−0.1130
Temperature (°F)	$8.544 \cdot 10^{-3}$
(Total wind + Crosswind) ² (mi ² /h ²)	$-3.204 \cdot 10^{-4}$
Denver*	0.3555
Playoff*	−0.4679
Defense quality (points/game allowed)	0.03282
Replacement kicker*	−0.5864
Fatigue*	−0.8368

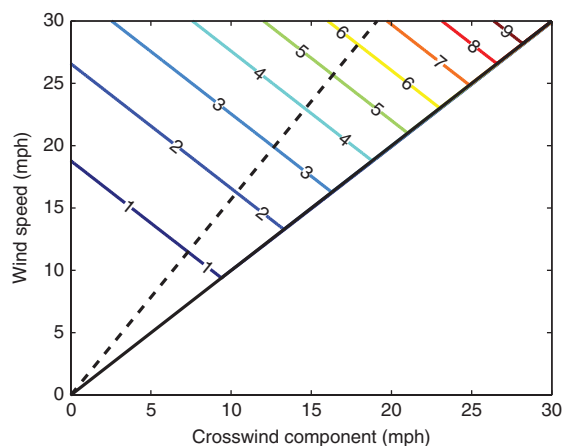
*Dimensionless binary variable.

Table 7 Approximate impact of various factors on effective distance.

Situation	Yards added
Playing in Denver (at altitude)	-3
Awful defense (allows 29 ppg)	-2
85°F temperature	-1
Bad defense (allows 25 ppg)	-1
10 mph diagonal wind	+1
Good defense (allows 19 ppg)	+1
45°F temperature	+2
Great defense (allows 15 ppg)	+2
15 mph crosswind	+3
20°F temperature	+4
Playoff game	+4
Backup kicker	+5
Fatigued kicker (5th FGA of game)	+7
25 mph crosswind	+7

ppg, points per game

then determine the effective yardage change associated with each. In such a model, we disregard the binary playoff, replacement kicker, and fatigue factors, as they are not associated with particular venues. Denver's altitude

**Figure 7** Contour plot of effective distance (in yards) added to a field goal attempt by wind. The dashed line indicates the best speed-only estimate for cases in which the wind direction is unknown. Note that the crosswind component of the wind speed cannot exceed the total wind speed.**Table 8** Easiest stadiums, by average yardage adjustment.

Stadium	Temp.	Wind	Altitude	Defense	Raw	Final
Denver	+1.47	+0.59	-3.15	-0.68	-1.77	-3.19
Detroit°	-	-	-	-0.89	-0.89	-2.31
Arizona°	-	-	-	-0.70	-0.70	-2.12
St. Louis°	-	-	-	-0.44	-0.44	-1.85
Atlanta°	-	-	-	+0.11	+0.11	-1.30

°Domed stadium.

Table 9 Toughest stadiums, by average yardage adjustment.

Stadium	Temp.	Wind	Altitude	Defense	Raw	Final
Buffalo	+2.11	+2.01	-	-0.13	+3.98	+2.57
Pittsburgh	+1.87	+1.04	-	+0.99	+3.90	+2.48
Cleveland	+1.98	+1.24	-	+0.17	+3.38	+1.97
Chicago	+1.93	+1.09	-	+0.15	+3.16	+1.75
Green Bay	+2.02	+0.74	-	+0.34	+3.10	+1.69

makes it the most kicker-friendly stadium, followed by domed stadiums with permanent (non-convertible) roofs, as shown in Table 8. The easiest outdoor stadiums are the three located in Florida. Predictably, the most difficult are northern stadiums in which cold, windy conditions frequently occur; these are listed in Table 9. The full results are given in Appendix C.

Returning to actual kicker performance, after accounting for the eight factors in our full model, the only stadiums in which visiting kickers' performance is significantly different from our expectations are San Diego (made 36/38, expected 31.9, $p < 0.05$) and Arizona (made 43/60, expected 51.1, $p < 0.005$). Arizona is theoretically among the easiest stadiums for kickers, yet (as shown in Table 2) had the second-lowest distance-adjusted success rate of any stadium in the league, hence the large disparity.

Under the NFL schedule format of this era, which is still in place as of the 2013 season, there were 256 regular-season games annually, plus 11 playoff contests. As shown in Table 10, seasonal league-wide success rates vary by a few percentage points, but expected success rates, based on our full model, were more consistent.

8 Rating kicker performance

League-wide expected success rates, based on a generic kicker, are a useful baseline when looking at the effects of individual stadiums or particular factors. However, to estimate a placekicker's contribution to his team, we need a different reference point, because kickers of comparable quality to the league average are fairly uncommon, and

Table 10 Seasonal league-wide summary statistics.

Season	FGA	FGM	FG%	ExpFG% [†]	Dist.*
2008	1038	877	84.5%	82.3%	35.2
2009	968	780	80.6%	82.7%	34.7
2010	994	818	82.3%	82.5%	34.8

[†]Based on the full eight-factor model, *Average distance of all made field goals.

Table 11 Top individual seasons, by PARK metric.

Player & Year	FGM/A	FG%	Repl%	FG%+	PARK	Dist.*
1) S. Janikowski '09	26/29	89.7% (14)	61.6% (1)	+28.1% (2)	+24.4 (1)	42.5 (2)
2) J. Hanson '08	21/22	95.5% (1)	64.8% (3)	+30.7% (1)	+20.3 (2)	42.9 (1)
3) R. Bironas '08	30/35	85.7% (32)	68.4% (8)	+17.3% (6)	+18.2 (3)	37.7 (11)
4) J. Reed '08	32/36	88.9% (18)	72.7% (30)	+16.2% (8)	+17.5 (4)	33.3 (70)
5) J. Brown '08	31/36	86.1% (31)	70.9% (16)	+15.2% (14)	+16.4 (5)	39.0 (8)
6) S. Gostkowski '08**	36/40	90.0% (12)	76.7% (59)	+13.3% (22)	+16.0 (6)	34.4 (56)
7) R. Bironas '09	27/32	84.4% (45)	68.4% (9)	+16.0% (9)	+15.3 (7)	39.1 (7)
8) A. Vinatieri '10	29/31	93.5% (4)	77.2% (64)	+16.4% (7)	+15.2 (8)	35.9 (30)
9) J. Kasay '08	28/31	90.3% (10)	74.5% (43)	+15.8% (10)	+14.7 (9)	36.7 (22)
10) J. Carney '08#	38/43	88.4% (22)	77.1% (63)	+11.3% (35)	+14.6 (10)	32.3 (86)

Rank, among 94 kicker-seasons with 16+ FGA, is in parentheses, *Average distance of all made field goals. #Pro Bowl selection, *1st team All-Pro (Associated Press), *2nd team All-Pro.

relatively expensive. If a team loses the starting kicker to injury, immediately signing another kicker of NFL-average ability is unlikely to occur. Instead, they would likely have a replacement-level kicker (as discussed in Sections 1 and 6) to fill in, so this is a more appropriate baseline against which to compare kickers. We define replacement FG% (denoted Repl%) as the expected field goal percentage for a replacement-level kicker, on some kick or set of kicks, according to our full, eight-factor model (with the binary replacement factor always set to one). In equations (4) and (5), we define field goal percentage added (FG%+) and points above replacement kicker (PARK), which are per-kick and sum total measures of a kicker's contribution, respectively. The coefficient of 3 in equation (5) comes from the number of points awarded to a team for a successful field goal. Note that PARK could alternatively be computed as 3 times the difference between the number of field goals made and the number expected to be made by a replacement kicker.

$$FG\%+ = FG\% - Repl\% \quad (4)$$

$$PARK = 3 \cdot FGA \cdot (FG\% - Repl\%) \quad (5)$$

The PARK measure favors those kickers who had more opportunities, as it scales with the number of attempts. This is intuitive; the (positive or negative) value to a team of a particular player is roughly linearly proportional to his playing time, at least up to the point where fatigue causes an impact. However, if a player's only kick of the season was a successful 50-yard attempt, then he would have the league's highest FG%+, if we did not restrict it based on the number of attempts. (Similarly, Major League Baseball has rules about the minimum number of plate appearances needed to qualify as the "batting champion.") In Tables 11 and 12, we list the top ten kickers by PARK and FG%+, respectively, from the set of 94 kicker-seasons with at least 16 FGA. The correlation between our two metrics is 0.94 (with each correlating to FG% at approximately 0.8), so it is not shocking that seven kicker-seasons appear on both lists.

For comparative purposes, Table 13 lists the kickers who were selected for the Pro Bowl (the NFL's

Table 12 Top individual seasons, by FG%+ metric.

Player & Year	FGM/A	FG%	Repl%	FG%+	PARK	Dist.*
1) J. Hanson '08	21/22	95.5% (1)	64.8% (3)	+30.7% (1)	+20.3 (2)	42.9 (1)
2) S. Janikowski '09	26/29	89.7% (14)	61.6% (1)	+28.1% (2)	+24.4 (1)	42.5 (2)
3) N. Folk '08	20/22	90.9% (8)	69.3% (11)	+21.6% (3)	+14.3 (11)	40.8 (4)
4) M. Prater '10	16/18	88.9% (18)	71.1% (18)	+17.8% (4)	+9.6 (36)	38.0 (10)
5) S. Suisham '10	17/20	85.0% (39)	67.5% (7)	+17.5% (5)	+10.5 (32)	34.9 (45)
6) R. Bironas '08	30/35	85.7% (32)	68.4% (8)	+17.3% (6)	+18.2 (3)	37.7 (11)
7) A. Vinatieri '10	29/31	93.5% (4)	77.2% (64)	+16.4% (7)	+15.2 (8)	35.9 (30)
8) J. Reed '08	32/36	88.9% (18)	72.7% (30)	+16.2% (8)	+17.5 (4)	33.3 (70)
9) R. Bironas '09	27/32	84.4% (45)	68.4% (9)	+16.0% (9)	+15.3 (7)	39.1 (7)
10) J. Kasay '08	28/31	90.3% (10)	74.5% (43)	+15.8% (10)	+14.7 (9)	36.7 (22)

Rank, among 94 kicker-seasons with 16+ FGA, is in parentheses, *Average distance of all made field goals. Pro Bowl selection, 1st team All-Pro (Associated Press), 2nd team All-Pro.

Table 13 Pro Bowl selections.

Player & Year	FGM/A	FG%	Repl%	FG%+	PARK	Dist.*
S. Gostkowski '08 [†]	36/40	90.0% (12)	76.7% (59)	+13.3% (22)	+16.0 (6)	34.4 (56)
J. Carney '08 [†]	38/43	88.4% (22)	77.1% (63)	+11.3% (35)	+14.6 (10)	32.3 (86)
N. Kaeding '09 [†]	32/38	84.2% (47)	77.7% (68)	+6.5% (56)	+7.4 (47)	32.2 (87)
D. Akers '09 [‡]	32/37	86.5% (28)	74.6% (44)	+11.8% (29)	+13.1 (14)	34.8 (46)
B. Cundiff '10 [†]	30/33	90.9% (8)	79.2% (78)	+11.7% (30)	+11.6 (26)	32.5 (80)
D. Akers '10 [‡]	33/41	80.5% (65)	74.1% (39)	+6.4% (59)	+7.9 (43)	34.0 (60)

Rank, among 94 kicker-seasons with 16+ FGA, is in parentheses, *Average distance of all made field goals. [†]1st team All-Pro (Associated Press), [‡]2nd team All-Pro

post-season “all-star” competition) over the three seasons; the kickers on the 1st team and 2nd team Associated Press All Pro teams for those years were the same six individuals, so this appears to show a popular consensus of the league’s top players at the position for those years. In comparing this group to the others, we note that the Pro Bowl kickers were among the leaders in field goal attempts and converted a good (but not great) percentage of their kicks, putting them near the top of the field goals made list. However, their kicks were relatively short and not particularly difficult, so their FG%+ ratings are unimpressive; none of this group was in the top 20% by the FG%+ metric. However, they fared a bit better using the PARK metric, aided by large numbers of attempts.

It is also notable that each of the six Pro Bowl kickers played on a team with a record of 10-6 or better; all but one of these teams made the playoffs. By comparison, the top FG%+ season (in Table 12) was achieved by Jason Hanson, whose 2008 Detroit Lions carry the dubious distinction of being the only 0-16 team in NFL history; given the Lions’ anemic offense, Hanson had relatively few field goal opportunities. When Sebastian Janikowski had a +24.4 PARK in 2009 (leading Table 11), his Oakland Raiders finished 5-11. Based on our sample, Pro Bowl selections seem to favor those who make a lot of short kicks (at a decent percentage) for a top team, as opposed to other kickers who may contribute more to weaker teams by converting more difficult field goals, but have fewer attempts. This is not so different than the typical measure of a pro basketball player, in which points scored per game (rather than shooting percentage or efficiency), is popularly considered the mark of an all-star player. During the 2012–2013 NBA season, nine of the eventual top ten per-game scoring leaders made the All-Star teams, compared to only three of the top ten in effective field goal percentage.

Among the starting kicker seasons, there is a correlation of 0.35 between the average distance of field goal attempts and the FG%+ statistic, as shown in Figure 8.

The positive correlation may be due to a combination of two issues: coaches’ confidence and the value of a starting kicker on long field goals. First, coaches may be more willing to try long field goals if they have an outstanding kicker, so the best kickers (who would have high FG%+ ratings) may get more long attempts, and thus a greater average attempt distance. However, as shown in Table 3 and Figure 6, the difference between starting and replacement kickers is more evident at longer distances. Because FG%+ is tied to the performance level of replacement kickers, an average starting kicker who gets more long attempts would expect to have a higher FG%+ than one who takes more short kicks.

In Figure 9, we attempt to illuminate the relationship between our two metrics and the number of attempts, by (with contour lines) showing the expected PARK value associated with a given number of attempts and particular field goal percentage, if the aggregate difficulty of those attempts was average. A replacement kicker would be expected to convert 75.0% of his kicks, if they were overall of average difficulty, explaining the placement

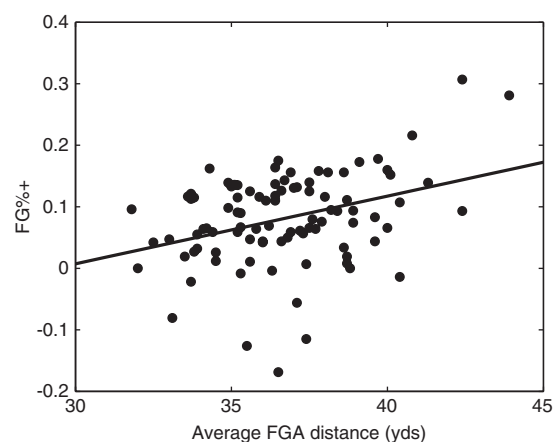


Figure 8 Scatter plot of single-season FG%+ values and average field goal attempt distances, for starting kickers. The best-fit regression line is shown.

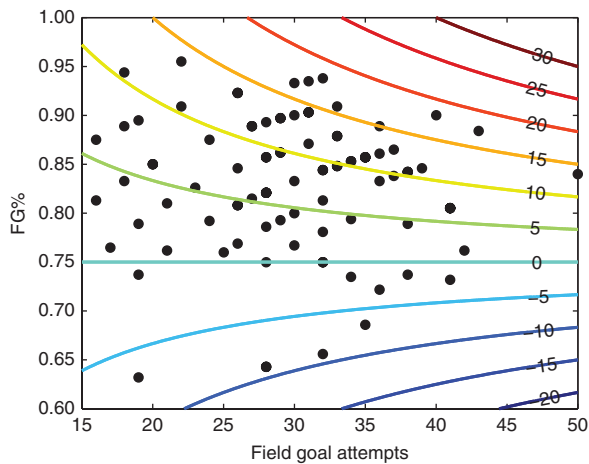


Figure 9 Expected PARK value (contour lines) for an average-difficulty set of kicks, with the given number of attempts (FGA) and success rate (FG%). Dots indicate FGA and FG%+ of starting kickers.

of the zero-valued contour line. The dots represent the single-season FGA and FG% of all starting kickers, but the actual PARK may have been higher or lower, based on the difficulty of a particular kicker's set of attempts. For example, Sebastian Janikowski's +24.4 PARK in 2009 exceeds the value for any dot on this figure, because he had the toughest slate of kicks of any kicker-season in the data set.

Over the three-season period, the top kickers by PARK were Tennessee's Rob Bironas (+44.6 expected points) and Oakland's Sebastian Janikowski (+43.1), both of whom were highly-ranked in the difficulty of kicks attempted. The top ten are shown in Table 14, and the results of all 35 kickers with 48+ FGA (averaging one attempt per regular season game) are included as Appendix A. Jason Hanson had the highest FG%+, but ranked only 4th in PARK, because he had far fewer attempts than those ahead of him.

9 Persistence of kicking skill

One way to differentiate between performance measures which are merely descriptive and those which are predictive is to consider the degree of correlation from one season to the next in that category. For example, in Major League Baseball, the slugging percentage (SLG) and on-base percentage (OBP) of batters correlate more highly across consecutive seasons than do hitter batting averages. This suggests that SLG and OBP are better predictors of future performance, and perhaps also of underlying hitting skill (Petti 2011).

Finding consistently-reliable kickers is a long-standing quandary for NFL coaches and front-office personnel, as discussed anecdotally by King (1992). A previous study found that, "[T]here is effectively no correlation between a kicker's field-goal percentage one season and his field-goal percentage the next (Schatz 2006)." In our data set, there were 53 cases in which a kicker had at least 16 FGA in each of two consecutive seasons, giving us a basis for computing season-to-season correlations of various kicking statistics. The results, shown in Table 15, support the conclusion of Schatz.

If some kickers are truly more skilled than others, we would expect to see positive correlations, indicating that high-performing kickers in one season are more likely to kick well again in the future, but this is not what our results show. Additionally, correlations by our two metrics of starting kicker performance from one season to the next are quite weak, with correlation coefficients of 0.01 for PARK and -0.10 for FG%+. The lack of an effect here does not necessarily mean that full-time NFL kickers are all equally skilled. Along the lines of argument made by Schuckers (2010) in studying hockey shootouts, it may be the case instead that true skill differentials are being masked by a high degree of randomness inherent

Table 14 Top kickers over the seasons 2008–2010, by PARK metric.

Player	FGM/A	FG%	Repl%	FG%+	PARK	Dist.*
1) R. Bironas	81/93	87.1% (4)	71.1% (4)	+16.0% (2)	+44.6 (1)	37.6 (3)
2) S. Janikowski	83/100	83.0% (24)	68.6% (2)	+14.4% (3)	+43.1 (2)	37.3 (4)
3) D. Akers	107/128	83.6% (23)	74.6% (16)	+9.0% (15)	+34.5 (3)	34.1 (24)
4) J. Hanson	54/64	84.4% (15)	66.7% (1)	+17.7% (1)	+34.0 (4)	40.2 (1)
5) J. Kasay	75/87	86.2% (8)	73.5% (9)	+12.7% (4)	+33.1 (5)	36.4 (7)
6) J. Brown	83/99	83.8% (20)	74.1% (12)	+9.8% (11)	+29.1 (6)	37.6 (2)
7) R. Gould	75/87	86.2% (8)	75.2% (18)	+11.0% (7)	+28.8 (7)	34.9 (14)
8) J. Feely	81/96	84.4% (15)	74.6% (15)	+9.8% (10)	+28.3 (8)	35.4 (13)
9) R. Longwell	74/82	90.2% (1)	78.9% (30)	+11.4% (5)	+28.0 (9)	34.3 (22)
10) N. Rackers	74/85	87.1% (5)	76.2% (21)	+10.8% (8)	+27.6 (10)	34.9 (15)

Rank, among 35 kickers with 48+ FGA, is in parentheses, *Average distance of all made field goals.

Table 15 Season-to-season correlations of kicking statistics.

Category	Correlation
Avg. FGA distance	0.42
Avg. FGM distance	0.29
Repl%	0.24
PARK	0.01
FG%	-0.02
FGA	-0.08
FG%+	-0.10
FGM	-0.11

in the phenomenon being studied, making it difficult to ascertain true skill. The relatively small sample sizes (number of attempts by each kicker per season, number of kickers in the data set, and number of seasons) could be an issue.

However, measures of the aggregate difficulty of a kicker's set of attempts do exhibit a positive correlation. This may relate to underlying leg strength differences among kickers; statistics on kickoff distance are highly correlated from one season to the next (Schatz 2006). Our results may also indicate that coaches' confidence in their kicker attempting long or difficult field goals is fairly stable across seasons, as might be expected over long careers (with some kickers often playing past age 40).

Comparing player performance to salaries can shed light on market perceptions of the relative value of certain skills. In an attempt to foster competitive balance, the NFL has a hard salary cap, i.e., a limit on the total amount that each team can pay its players; this prevents wealthier teams/owners from signing the bulk of top-tier free agents. Because of the complex nature of some professional

athletes' contracts, which may include signing bonuses, performance incentives, and other types of non-guaranteed compensation, we will use the "cap hit" (the amount counted toward a team's total) as our salary number for each player. A full list of salaries (*USA Today* 2010) was published by *USA Today* at one time, but this was discontinued after the 2009 season, so we do not have 2010 salary data.

Figure 10 is a scatter plot of single-season PARK values and salaries (in millions), for starting kickers during the 2008 and 2009 seasons. The correlation between these variables is just 0.12 (and the correlation between salary and FG%+ is similar, at 0.11). The regression line has a slope of just 1.07, indicating that spending one million dollars more on kicker salary would be expected to yield (on average) an additional 1.07 points on field goals. This is in line with the findings of Schlauch (2010). For an average team, it takes about additional 35 points scored and/or prevented to add one win (Meers 2012), and even more for a team near the top or bottom of the standings. Given an annual team salary cap on the order of \$115–\$130 million (Gaines 2011) for a 53-man roster, this makes elite field goal kickers expensive, in terms of return on investment. It is notable that our data indicates that a kicker with a near-minimum salary would be expected to have a PARK of about +6 points (as compared to zero for a replacement player); this is likely indicative of survivorship bias, as replacement-level kickers rarely retain starting positions long enough to get 16 attempts in a season. If we remove kickers with salaries under \$1 million, so that all kickers in the salary sample were expected by their teams to be starters, then the best-fit line has a negative slope, indicating a lower PARK value from higher-paid kickers.

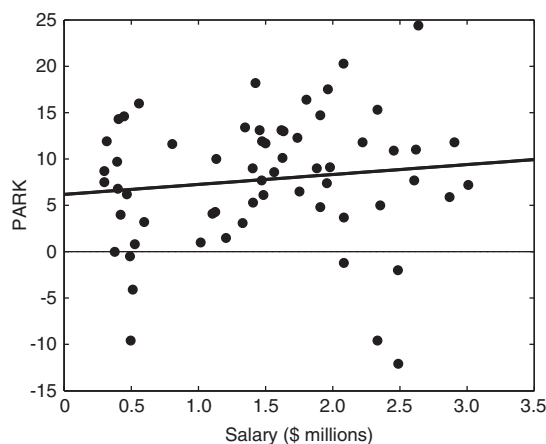


Figure 10 Scatter plot of single-season PARK values and salaries, for 2008 and 2009 starting kickers. The best-fit regression line has equation $PARK = 6.18 + 1.07 \cdot \text{salary}$.

10 Future work

In settling on a logistic regression model, for reasons of clarity and simplicity, we must acknowledge its inherent limitations. Notably, the distance/probability curve has an inflection point where $P=0.5$ (at about 55 yards for our distance-only model), so our models overestimate the likelihood of very long field goals being converted. For example, Sebastian Janikowski's 76-yard attempt in 2008 (which was far short) is given a success probability of 6.1%, even though this was 13 yards longer than the NFL record, and occurred in a stiff wind.

Limited samples under certain conditions (such as 99 attempts at Denver's high altitude, 106 playoff attempts, 142 under pressure, etc.) reduce our confidence in the

parameter estimates associated with some phenomena included in the model. Expanding the data set to a decade's worth of kicks would help alleviate these issues. We did not consider effects of gusty (as opposed to strong, but sustained) winds, but they may be significant. Effects of psychological pressure late in a close contest and the practice of "icing" a kicker by calling a defensive timeout (both studied in Berry and Wood 2004) might also be worth revisiting.

If others find our model to be too complex, and question the relevance of some of the factors, a simple distance-only model would also provide a useful efficiency metric, far superior to field goal percentage. In the set of 35 kickers who had 48+ FGA over the three seasons (see Section 8 and Appendix A), the correlation between PARK and a PARK-equivalent metric based only distance and replacement is 0.95; similarly, FG%+ and its' distance-based analogous measure correlate at 0.94.

Our metric does not take into account the value of gaining or losing field position due to converting kicks or failing to do so, but Romer point values (Romer 2006) could be used for this, given that our PARK metric has the same unit (points). More importantly, in rating kickers, we have not considered the impact of kickoffs. The ability of a placekicker to routinely force touchbacks might contribute as much or more value to his team as the ability to accurately kick field goals. We have intentionally omitted point-after touchdown (PAT) attempts, due to the rarity of failure, as nearly 99% of all PAT attempts over the last decade have succeeded, according to data at Pro Football Reference.

Finally, some in-game decision theory analysis (on 4th down or the closing seconds of a half, choosing whether to punt, to attempt a field goal, or to run an offensive play) may be advanced in light of a field goal probability model. This would be particularly significant if a more sophisticated model found useful information about the relative leg strength and accuracy of individual kickers.

Appendix A Overall Kicking Results, by PARK (minimum 48 FGA)

Player & Year	FGM/A	FG%	Repl%	FG%+	PARK	FGM dist.
1) R. Bironas	81/93	87.1% (4)	71.1% (4)	+16.0% (2)	+44.6 (1)	37.6 (3)
2) S. Janikowski	83/100	83.0% (24)	68.6% (2)	+14.4% (3)	+43.1 (2)	37.3 (4)
3) D. Akers	107/128	83.6% (23)	74.6% (16)	+9.0% (15)	+34.5 (3)	34.1 (24)
4) J. Hanson	54/64	84.4% (15)	66.7% (1)	+17.7% (1)	+34.0 (4)	40.2 (1)
5) J. Kasay	75/87	86.2% (8)	73.5% (9)	+12.7% (4)	+33.1 (5)	36.4 (7)
6) J. Brown	83/99	83.8% (20)	74.1% (12)	+9.8% (11)	+29.1 (6)	37.6 (2)
7) R. Gould	75/87	86.2% (8)	75.2% (18)	+11.0% (7)	+28.8 (7)	34.9 (14)
8) J. Feely	81/96	84.4% (15)	74.6% (15)	+9.8% (10)	+28.3 (8)	35.4 (13)
9) R. Longwell	74/82	90.2% (1)	78.9% (30)	+11.4% (5)	+28.0 (9)	34.3 (22)
10) N. Rackers	74/85	87.1% (5)	76.2% (21)	+10.8% (8)	+27.6 (10)	34.9 (15)
11) D. Carpenter	77/95	81.1% (28)	71.8% (5)	+9.3% (13)	+26.5 (11)	36.4 (6)
12) J. Reed	83/99	83.8% (20)	75.1% (17)	+8.7% (17)	+25.9 (12)	33.0 (29)
13) O. Mare	76/86	88.4% (2)	79.4% (31)	+9.0% (14)	+23.2 (13)	33.7 (26)
14) J. Nedney	57/67	85.1% (11)	73.8% (10)	+11.3% (6)	+22.7 (14)	35.8 (10)
15) P. Dawson	70/83	84.3% (17)	75.5% (19)	+8.8% (16)	+21.9 (15)	32.9 (30)
16) R. Lindell	74/92	80.4% (29)	72.7% (6)	+7.7% (20)	+21.2 (16)	34.5 (19)
17) N. Kaeding	84/100	84.0% (18)	77.3% (25)	+6.7% (22)	+20.1 (17)	33.5 (28)
18) S. Gostkowski	72/85	84.7% (14)	76.9% (23)	+7.8% (19)	+19.9 (18)	34.3 (21)
19) A. Vinatieri	57/66	86.4% (7)	76.5% (22)	+9.8% (9)	+19.5 (19)	35.8 (9)
20) G. Hartley	50/57	87.7% (3)	78.2% (28)	+9.6% (12)	+16.4 (20)	34.4 (20)
21) S. Graham	58/68	85.3% (10)	77.4% (26)	+7.9% (18)	+16.2 (21)	32.6 (32)
22) M. Prater	71/87	81.6% (27)	75.6% (20)	+6.1% (25)	+15.8 (22)	35.7 (12)
23) M. Bryant	67/79	84.8% (13)	78.3% (29)	+6.5% (24)	+15.4 (23)	33.9 (25)
24) R. Succop	45/55	81.8% (26)	74.2% (14)	+7.6% (21)	+12.5 (24)	34.5 (18)
25) S. Suisham	66/85	77.6% (31)	72.8% (7)	+4.9% (28)	+12.4 (25)	34.8 (16)
26) M. Crosby	80/104	76.9% (33)	73.0% (8)	+3.9% (30)	+12.1 (26)	34.1 (23)
27) B. Cundiff	51/59	86.4% (6)	79.9% (32)	+6.5% (23)	+11.6 (27)	31.4 (35)
28) C. Barth	47/59	79.7% (30)	74.1% (13)	+5.5% (27)	+9.8 (28)	35.9 (8)
29) M. Stover	46/55	83.6% (22)	78.0% (27)	+5.6% (26)	+9.3 (29)	31.9 (33)
30) J. Carney	56/66	84.8% (12)	80.4% (34)	+4.5% (29)	+8.9 (30)	31.9 (34)

(Appendix A Continued)

Player & Year	FGM/A	FG%	Repl%	FG%+	PARK	FGM dist.
31) J. Scobee	59/81	72.8% (35)	69.8% (3)	+3.0% (31)	+7.3 (31)	37.3 (5)
32) N. Folk	70/92	76.1% (34)	73.9% (11)	+2.2% (34)	+6.0 (32)	35.7 (11)
33) L. Tynes	47/56	83.9% (19)	81.2% (35)	+2.8% (32)	+4.6 (33)	32.7 (31)
34) J. Elam	42/51	82.4% (25)	80.1% (33)	+2.2% (33)	+3.4 (34)	34.6 (17)
35) K. Brown	54/70	77.1% (32)	77.0% (24)	+0.2% (35)	+0.3 (35)	33.6 (27)

Appendix B Seasonal Kicking Results, by PARK (minimum 16 FGA)

Player & Year	FGM/A	FG%	Repl%	FG%+	PARK	FGM dist.
1) S. Janikowski '09	26/29	89.7% (14)	61.6% (1)	+28.1% (2)	+24.4 (1)	42.5 (2)
2) J. Hanson '08	21/22	95.5% (1)	64.8% (3)	+30.7% (1)	+20.3 (2)	42.9 (1)
3) R. Bironas '08	30/35	85.7% (32)	68.4% (8)	+17.3% (6)	+18.2 (3)	37.7 (11)
4) J. Reed '08	32/36	88.9% (18)	72.7% (30)	+16.2% (8)	+17.5 (4)	33.3 (70)
5) J. Brown '08	31/36	86.1% (31)	70.9% (16)	+15.2% (14)	+16.4 (5)	39.0 (8)
6) S. Gostkowski '08	36/40	90.0% (12)	76.7% (59)	+13.3% (22)	+16.0 (6)	34.4 (56)
7) R. Bironas '09	27/32	84.4% (45)	68.4% (9)	+16.0% (9)	+15.3 (7)	39.1 (7)
8) A. Vinatieri '10	29/31	93.5% (4)	77.2% (64)	+16.4% (7)	+15.2 (8)	35.9 (30)
9) J. Kasay '08	28/31	90.3% (10)	74.5% (43)	+15.8% (10)	+14.7 (9)	36.7 (22)
10) J. Carney '08	38/43	88.4% (22)	77.1% (63)	+11.3% (35)	+14.6 (10)	32.3 (86)
11) N. Folk '08	20/22	90.9% (8)	69.3% (11)	+21.6% (3)	+14.3 (11)	40.8 (4)
12) J. Kasay '10	25/29	86.2% (29)	70.6% (14)	+15.6% (11)	+13.6 (12)	37.6 (12)
13) D. Akers '08	42/50	84.0% (49)	75.0% (48)	+9.0% (47)	+13.4 (13)	33.6 (66)
14) D. Akers '09	32/37	86.5% (28)	74.6% (44)	+11.8% (29)	+13.1 (14)	34.8 (46)
15) J. Elam '08	30/32	93.8% (3)	80.1% (84)	+13.6% (20)	+13.1 (15)	34.6 (50)
16) J. Nedney '08	29/33	87.9% (23)	74.8% (47)	+13.1% (24)	+13.0 (16)	35.7 (35)
17) J. Feely '10	24/27	88.9% (18)	73.3% (32)	+15.6% (12)	+12.6 (17)	36.6 (26)
18) K. Brown '08	29/33	87.9% (23)	75.4% (50)	+12.5% (26)	+12.3 (18)	34.6 (51)
19) R. Succop '09	25/29	86.2% (29)	72.5% (29)	+13.7% (19)	+11.9 (19)	34.0 (59)
20) P. Dawson '08	30/36	83.3% (51)	72.3% (28)	+11.0% (37)	+11.9 (20)	34.6 (52)
21) N. Rackers '10	27/30	90.0% (12)	76.8% (60)	+13.2% (23)	+11.9 (21)	35.9 (31)
22) R. Longwell '08	29/34	85.3% (37)	73.7% (36)	+11.6% (31)	+11.8 (22)	37.1 (17)
23) R. Gould '09	24/28	85.7% (32)	71.7% (24)	+14.0% (16)	+11.8 (23)	35.7 (33)
24) J. Feely '09	33/41	80.5% (65)	71.0% (17)	+9.5% (42)	+11.7 (24)	36.8 (19)
25) N. Rackers '08	30/35	85.7% (32)	74.7% (45)	+11.0% (38)	+11.6 (25)	34.0 (60)
26) B. Cundiff '10	30/33	90.9% (8)	79.2% (78)	+11.7% (30)	+11.6 (26)	32.5 (80)
27) S. Janikowski '10	33/41	80.5% (65)	71.2% (19)	+9.3% (45)	+11.4 (27)	36.3 (28)
28) R. Bironas '10	24/26	92.3% (6)	78.1% (70)	+14.3% (15)	+11.1 (28)	35.7 (33)
29) R. Gould '08	26/29	89.7% (14)	77.0% (62)	+12.6% (25)	+11.0 (29)	35.5 (37)
30) R. Longwell '09	28/30	93.3% (5)	81.2% (92)	+12.1% (28)	+10.9 (30)	33.1 (74)
31) M. Bryant '10	28/31	90.3% (10)	78.8% (75)	+11.5% (34)	+10.7 (31)	34.5 (54)
32) S. Suisham '10	17/20	85.0% (39)	67.5% (7)	+17.5% (5)	+10.5 (32)	34.9 (45)
33) O. Mare '08	24/27	88.9% (18)	76.4% (57)	+12.5% (27)	+10.1 (33)	36.7 (23)
34) S. Graham '08	21/24	87.5% (25)	73.6% (35)	+13.9% (17)	+10.0 (34)	33.6 (66)
35) D. Carpenter '09	25/28	89.3% (17)	77.7% (67)	+11.6% (32)	+9.7 (35)	35.2 (40)
36) M. Prater '10	16/18	88.9% (18)	71.1% (18)	+17.8% (4)	+9.6 (36)	38.0 (10)
37) J. Reed '09	27/31	87.1% (27)	77.3% (65)	+9.8% (40)	+9.1 (37)	33.2 (73)
38) R. Lindell '09	28/33	84.8% (41)	75.7% (52)	+9.1% (46)	+9.0 (38)	32.7 (78)
39) O. Mare '09	24/26	92.3% (6)	80.8% (88)	+11.5% (33)	+9.0 (39)	33.4 (69)
40) D. Carpenter '08	22/26	84.6% (43)	73.5% (33)	+11.1% (36)	+8.7 (40)	37.3 (14)
41) R. Lindell '08	30/38	78.9% (72)	71.4% (20)	+7.6% (50)	+8.6 (41)	36.4 (27)
42) D. Carpenter '10	30/41	73.2% (88)	66.6% (5)	+6.6% (55)	+8.1 (42)	36.8 (21)
43) D. Akers '10	33/41	80.5% (65)	74.1% (39)	+6.4% (59)	+7.9 (43)	34.0 (60)
44) P. Dawson '09	17/19	89.5% (16)	76.0% (55)	+13.5% (21)	+7.7 (44)	33.7 (65)
45) J. Brown '09	19/24	79.2% (71)	68.5% (10)	+10.7% (39)	+7.7 (45)	39.1 (6)
46) G. Hartley '09	14/16	87.5% (25)	71.9% (25)	+15.6% (13)	+7.5 (46)	35.4 (39)
47) N. Kaeding '09	32/38	84.2% (47)	77.7% (68)	+6.5% (56)	+7.4 (47)	32.2 (87)

(Appendix B Continued)

Player & Year	FGM/A	FG%	Repl%	FG%+	PARK	FGM dist.
48) S. Janikowski '08	24/30	80.0% (68)	72.0% (26)	+8.0% (49)	+7.2 (48)	33.0 (76)
49) J. Scobee '10	22/28	78.6% (74)	70.2% (12)	+8.3% (48)	+7.0 (49)	38.2 (9)
50) M. Crosby '08	27/34	79.4% (69)	72.8% (31)	+6.6% (54)	+6.8 (50)	35.1 (41)
51) D. Rayner '10	13/16	81.3% (60)	67.4% (6)	+13.9% (18)	+6.7 (51)	39.8 (5)
52) N. Kaeding '08	29/34	85.3% (37)	78.9% (76)	+6.4% (57)	+6.5 (52)	32.4 (82)
53) N. Kaeding '10	23/28	82.1% (55)	74.8% (46)	+7.4% (51)	+6.2 (53)	36.8 (20)
54) M. Prater '09	30/35	85.7% (32)	79.8% (82)	+5.9% (63)	+6.2 (54)	33.8 (64)
55) M. Stover '08	31/37	83.8% (50)	78.3% (74)	+5.5% (65)	+6.1 (55)	31.7 (88)
56) R. Gould '10	25/30	83.3% (51)	76.6% (58)	+6.7% (53)	+6.1 (56)	33.4 (68)
57) J. Nedney '09	17/21	81.0% (62)	71.6% (22)	+9.4% (43)	+5.9 (57)	37.2 (15)
58) M. Crosby '10	25/32	78.1% (75)	72.0% (27)	+6.1% (60)	+5.8 (58)	34.9 (43)
59) C. Barth '09	14/19	73.7% (85)	64.3% (2)	+9.3% (44)	+5.3 (59)	41.6 (3)
60) M. Bryant '08	32/38	84.2% (47)	79.5% (80)	+4.7% (68)	+5.3 (60)	33.2 (72)
61) R. Longwell '10	17/18	94.4% (2)	84.8% (94)	+9.6% (41)	+5.2 (61)	31.4 (91)
62) A. Vinatieri '08	21/26	80.8% (63)	74.4% (42)	+6.4% (58)	+5.0 (62)	37.2 (16)
63) J. Brown '10	33/39	84.6% (43)	80.4% (85)	+4.2% (72)	+4.9 (63)	35.5 (38)
64) J. Kasay '09	22/27	81.5% (59)	75.6% (51)	+5.9% (62)	+4.8 (64)	34.5 (55)
65) S. Gostkowski '09	26/32	81.3% (60)	76.8% (61)	+4.4% (69)	+4.3 (65)	34.5 (53)
66) N. Rackers '09	17/20	85.0% (39)	78.1% (71)	+6.9% (52)	+4.1 (66)	34.8 (48)
67) O. Mare '10	28/33	84.8% (41)	80.7% (87)	+4.2% (73)	+4.1 (67)	31.4 (89)
68) J. Feely '08	24/28	85.7% (32)	81.0% (90)	+4.7% (67)	+4.0 (68)	32.4 (83)
69) J. Hanson '09	21/28	75.0% (82)	70.6% (15)	+4.4% (70)	+3.7 (69)	36.6 (25)
70) C. Barth '10	23/28	82.1% (55)	77.8% (69)	+4.4% (71)	+3.7 (70)	34.9 (44)
71) R. Lindell '10	16/21	76.2% (79)	70.5% (13)	+5.7% (64)	+3.6 (71)	33.9 (62)
72) D. Beuhler '10	24/32	75.0% (82)	71.6% (23)	+3.4% (74)	+3.2 (72)	36.9 (18)
73) M. Stover '09	15/18	83.3% (51)	77.4% (66)	+5.9% (61)	+3.2 (73)	32.3 (85)
74) L. Tynes '09	27/32	84.4% (45)	81.1% (91)	+3.2% (75)	+3.1 (74)	34.1 (58)
75) M. Nugent '10	15/19	78.9% (72)	74.0% (37)	+5.0% (66)	+2.8 (75)	33.8 (63)
76) P. Dawson '10	23/28	82.1% (55)	79.4% (79)	+2.7% (76)	+2.3 (76)	30.2 (93)
77) G. Hartley '10	23/28	82.1% (55)	79.6% (81)	+2.6% (77)	+2.2 (77)	34.8 (47)
78) J. Scobee '08	19/25	76.0% (81)	74.1% (38)	+1.9% (78)	+1.5 (78)	37.6 (13)
79) N. Folk '10	32/42	76.2% (79)	75.1% (49)	+1.1% (81)	+1.3 (79)	33.3 (70)
80) L. Tynes '10	19/23	82.6% (54)	80.7% (86)	+1.9% (79)	+1.3 (80)	31.4 (90)
81) S. Suisham '09	23/29	79.3% (70)	78.1% (72)	+1.2% (80)	+1.0 (81)	33.1 (75)
82) S. Suisham '08	26/36	72.2% (89)	71.4% (21)	+0.8% (82)	+0.8 (82)	36.3 (29)
83) R. Succop '10	20/26	76.9% (76)	76.2% (56)	+0.7% (83)	+0.6 (83)	35.1 (42)
84) B. Cundiff '09	21/26	80.8% (63)	80.8% (89)	0.0% (84)	0.0 (84)	29.9 (94)
85) M. Prater '08	25/34	73.5% (87)	73.5% (34)	0.0% (85)	0.0 (85)	36.6 (24)
86) M. Crosby '09	28/38	73.7% (85)	74.1% (40)	-0.4% (86)	-0.5 (86)	32.4 (84)
87) J. Reed '10	24/32	75.0% (82)	75.8% (54)	-0.8% (87)	-0.7 (87)	32.5 (79)
88) J. Scobee '09	18/28	64.3% (92)	65.7% (4)	-1.4% (88)	-1.2 (88)	35.8 (32)
89) S. Graham '09	23/30	76.7% (77)	78.9% (77)	-2.2% (89)	-2.0 (89)	32.8 (77)
90) J. Carney '09	13/17	76.5% (78)	84.6% (93)	-8.1% (91)	-4.1 (90)	31.2 (92)
91) G. Gano '10	24/35	68.6% (90)	74.2% (41)	+5.6% (90)	-5.9 (91)	35.6 (36)
92) N. Folk '09	18/28	64.3% (92)	75.8% (53)	-11.5% (92)	-9.6 (92)	34.4 (56)
93) J. Elam '09	12/19	63.2% (94)	80.1% (83)	-16.9% (94)	-9.6 (93)	34.7 (49)
94) K. Brown '09	21/32	65.6% (91)	78.2% (73)	-12.6% (93)	-12.1 (94)	32.4 (81)

Appendix C Stadium difficulty, by average yardage adjustment. Raw yardage is relative to a domed stadium with an average defense. Final yardage is relative to the NFL league-wide average value.

Stadium	Temp.	Wind	Altitude	Defense	Raw	Final
Buffalo	+2.11	+2.01	–	–0.13	+3.98	+2.57
Pittsburgh	+1.87	+1.04	–	+0.99	+3.90	+2.48
Cleveland	+1.98	+1.24	–	+0.17	+3.38	+1.97
Chicago	+1.93	+1.09	–	+0.15	+3.16	+1.75
Green Bay	+2.02	+0.74	–	+0.34	+3.10	+1.69
Cincinnati	+1.25	+1.64	–	+0.11	+3.00	+1.59
Baltimore	+1.28	+0.49	–	+0.71	+2.48	+1.06
New England	+1.41	+0.68	–	+0.32	+2.41	+1.00
San Francisco	+0.67	+1.89	–	–0.16	+2.40	+0.98
Misc. venues	+2.22	+0.56	–	–0.41	+2.38	+0.97
New York (old)	+1.13	+0.96	–	+0.27	+2.36	+0.94
Philadelphia	+1.21	+0.79	–	+0.22	+2.22	+0.80
Kansas City	+1.31	+1.20	–	–0.66	+1.86	+0.44
Dallas (old)	+0.05	+1.82	–	–0.09	+1.78	+0.36
New York (new)	+1.22	+0.44	–	+0.10	+1.75	+0.34
Tennessee	+0.79	+0.58	–	+0.22	+1.59	+0.17
Oakland	+0.79	+1.02	–	–0.32	+1.50	+0.08
Washington	+0.98	+0.37	–	+0.05	+1.41	–0.01
Seattle	+1.33	+0.49	–	–0.68	+1.14	–0.27
Carolina	+0.86	+0.38	–	–0.14	+1.10	–0.32
San Diego	+0.25	+0.42	–	+0.17	+0.85	–0.57
Dallas (new) [†]	+0.07	+0.54	–	+0.14	+0.75	–0.67
Tampa Bay	–0.27	+0.77	–	+0.18	+0.69	–0.73
Indianapolis [†]	+0.22	+0.29	–	+0.15	+0.65	–0.76
Houston [†]	+0.17	+0.53	–	–0.10	+0.60	–0.81
Miami	–0.48	+0.81	–	+0.26	+0.59	–0.83
Jacksonville	+0.07	+0.80	–	–0.48	+0.38	–1.03
New Orleans [†]	–	–	–	+0.24	+0.24	–1.17
Minnesota [†]	–	–	–	+0.19	+0.19	–1.22
Atlanta [†]	–	–	–	+0.11	+0.11	–1.30
St. Louis [†]	–	–	–	–0.44	–0.44	–1.85
Arizona [†]	–	–	–	–0.70	–0.70	–2.12
Detroit [†]	–	–	–	–0.89	–0.89	–2.31
Denver	+1.47	+0.59	–3.15	–0.68	–1.77	–3.19

[†]Domed stadium with convertible roof (sometimes closed).

[‡]Domed stadium with permanent roof (always closed).

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